

Harvard Extension Data Science

Dynamic Modeling and Forecasting in Big Data

Instructor: William Yu

Final Project

- **Project Goal:** The objective of this final project is to apply time series and/or dynamic models to a real-world dataset and generate meaningful predictions.
- **Topic Selection:** You are responsible for choosing your project topic. Select one based on your personal interests, career goals, or relevant work experience.
- **Data Requirement:** You may use any dataset, and larger datasets are encouraged
 - The only requirement is that the dataset must include a time-related variable—such as year, month, day, or intra-day frequency.
 - If you're having trouble finding data, consider the following sources (additional links are provided on the next page):
 - <https://www.kaggle.com/datasets>
 - <https://github.com/awesomedata/awesome-public-datasets>
- You *cannot* submit a project that has already been completed or submitted for another class.
- Group project is allowed. Each participant will share the same grade.

Three Components of the Final Project

(1) PowerPoint Presentation (Optional for Undergraduate Credit)

- Prepare a slide deck (maximum 10 slides) and a 10-minute presentation. There are two options of presentation.
 - Live presentation on the *final instruction day*. Sign-up is required as only 10 slots are available. A participation bonus will be awarded for real-time presentations.
 - Pre-recorded presentation using Zoom or YouTube. Submit the recording along with your project materials.

(2) Written Report

- Your written report should include the following elements:
 - Project purpose and motivation: Explain the objective and significance of your analysis.
 - Visualizations: Present clear, high-quality graphs and charts.
 - Model selection and justification: Describe the models you used and explain why you chose them. Include test-set or out-of-sample validation.
 - Narrative and insights: Tell a compelling story and summarize your key findings and takeaways.
- Length: 5–20 pages (including charts and figures).

(3) Codes: R Script, R Markdown, or Python Jupyter Notebook

- **Submission Details:** Submit all project materials (slides, report, code, and dataset) via Canvas by midnight, **December 18**.

- Your submission will be shared with the class on Canvas for educational purposes. If your data or report is proprietary and cannot be shared, please inform me in your report. I will not share it to the class. If your dataset is big, such as bigger than 50MB, you don't need to submit it. Just reveal the source or its website link.
- Grading will be based primarily on the four components outlined in the written report and presentation. Projects that stand out typically feature: (1) A compelling and well-structured story telling. (2) Robust and thoughtful data analysis. (3) A novel or creative approach in topic, methodology, or application.

Useful Links and Public Datasets

- <https://ourworldindata.org/>
- <https://www.kaggle.com/competitions> (Kaggle competitions)
- <https://www.dataquest.io/blog/free-datasets-for-projects/>
- <http://archive.ics.uci.edu/ml/index.php> (UCI Machine Learning Repository)
- <https://www.kdnuggets.com/> (KD Nuggets)
- <https://stackoverflow.com/tags> (Stack Overflow)
- <https://stats.stackexchange.com/tags> (Cross Validated)
- Federal Reserve Economic Data (FRED, <https://research.stlouisfed.org/fred2/>)
- Bureau of Economic Analysis (BEA, <http://www.bea.gov/>)
- Bureau of Labor Statistics (BLS, <http://www.bls.gov/>)
- Google Public Data Explorer with dynamic graphing.
- Yahoo Finance (<http://finance.yahoo.com/>), providing downloadable historical finance data
- Fed Regional Banks' data, indicators, and research: New York, San Francisco, Chicago, Kansas City, Dallas, Philadelphia, Cleveland, Atlanta, Boston, Minneapolis, Richmond, DC Board
- Gapminder, a fact-based worldview, with a very cool chart!
- Visualizing Economics, providing a lot of vivid charts and maps in the world!
- World Bank's World Development Indicators and Worldwide Governance Indicators
- International Monetary Fund (IMF)'s World Economic Outlook Databases
- National Bureau of Economic Research (NBER)'s data, collecting a bunch of great research data!
- International regional and country's statistical sites by BLS and by EDIRC
- Index of Economic Freedom and CIA the World Factbook
- Robert Shiller's stock and housing markets data, providing useful financial market data
- S&P/Case-Shiller home price indices
- Federal Housing Finance Agency (FHFA)'s House Price Index
- Zillow Real Estate Research data (<http://www.zillow.com/research/data/>)
- IMF Global Housing Watch and Dallas Fed's International House Price Database
- BIS Property Price Statistics
- Angus Maddison data, two thousand year GDP per capita data around the world.
- Citylab's Maps (<http://www.citylab.com/posts/maps/>)
- Google Books Ngram Viewer (word frequency in books from 1800 to date)
- The Mnist database (<http://yann.lecun.com/exdb/mnist/>)
- Image Net (<http://image-net.org/>)
- Word Net (<https://wordnet.princeton.edu/>)
- Visual Genome (<http://visualgenome.org/>)